LECTURE 8: SEQUENTIAL GAMES

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WEEK 3

LETTING SOMEONE GO FIRST

- Up to now, we've thought of games played at the same moment
- But now we'll think of different settings where one player goes first, and another player has to respond

  - First firm to enter a market might get higher market share

  - In tennis, the server gets to dictate how the first part of the rally is played

  - In chess, white gets to influence how the game will play out

- Whoever goes first might get an **advantage** because they can force the hand of the other play

LETTING SOMEONE GO FIRST

- In simultaneous games, each player has a best response to the other player
- In sequential games, only the second mover responds to the first
- The first mover can then predict what the second mover will do, and then choose its action based on that (backwards induction or backsolving)

LET'S GO BACK TO OUR SIMPLE ENTRY GAME

There a two firms looking to enter a market. If both enter, they both make $-1 profit, if only one enters then that firm makes $1 profit, otherwise they both make no money

	Firm 1 enters	Firm 1 does not enter
Firm 2 enters	F1 profit = -1 F2 profit = -1	F1 profit = 0 F2 profit = 1
Firm 2 does not enter	F1 profit = 1 F2 profit = 0	F1 profit = 0 F2 profit = 0

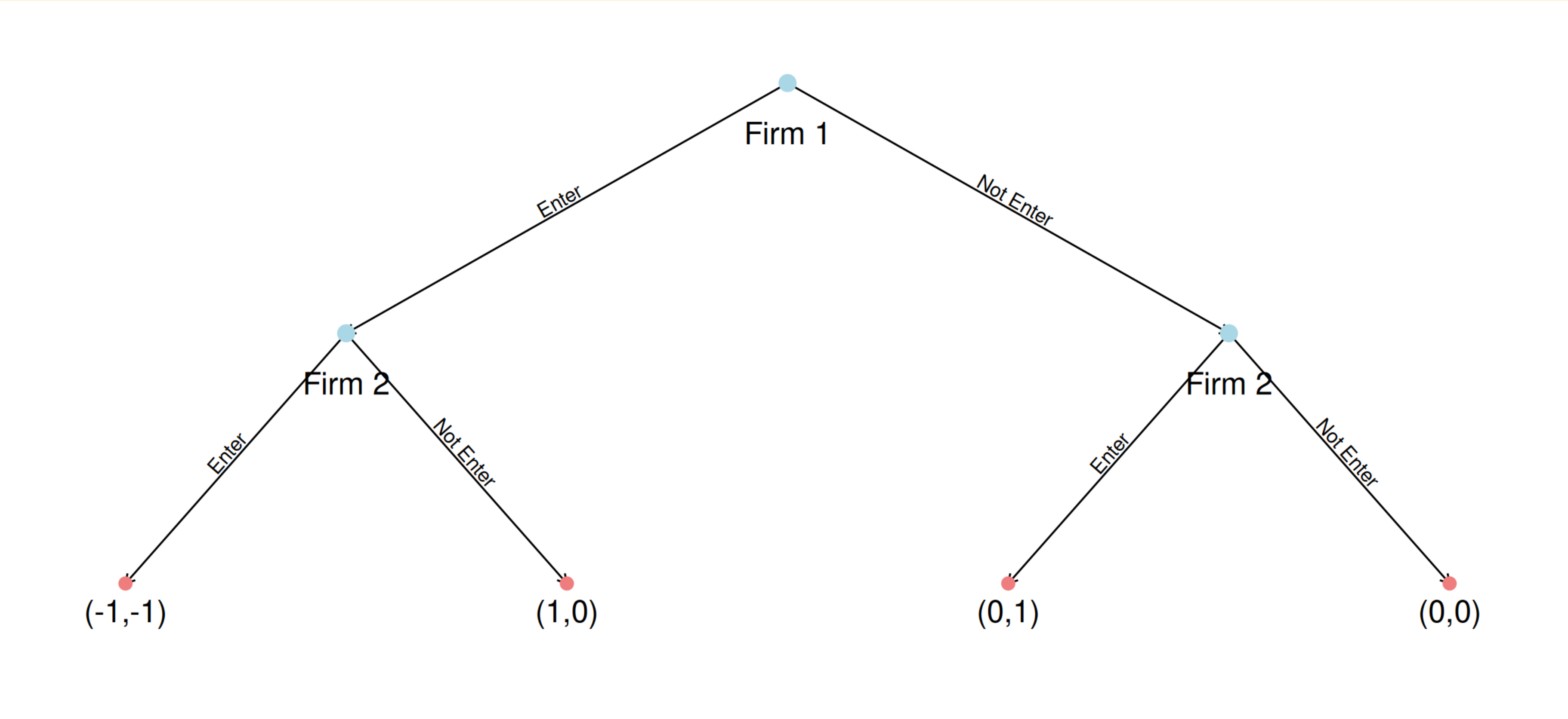
- We showed in the last lecture, than when this is played simultaneously, each firm chooses with 50% probability to enter or not
- This is because the firms make simultaneous decisions, so neither knows what the other is going to do

NOW LET'S PLAY THE FIRM ENTRY GAME AS A SEQUENTIAL GAME

- But what if Firm 1 is quicker than Firm 2?
- If Firm 1 can choose before Firm 2 chooses - does it get an advantage in entering the market?
- We want to think about how Firm 1 choosing first changes the game outcome

NOW LET'S PLAY THE FIRM ENTRY GAME AS A SEQUENTIAL GAME

We draw this out in what's called a game tree - or in extensive form

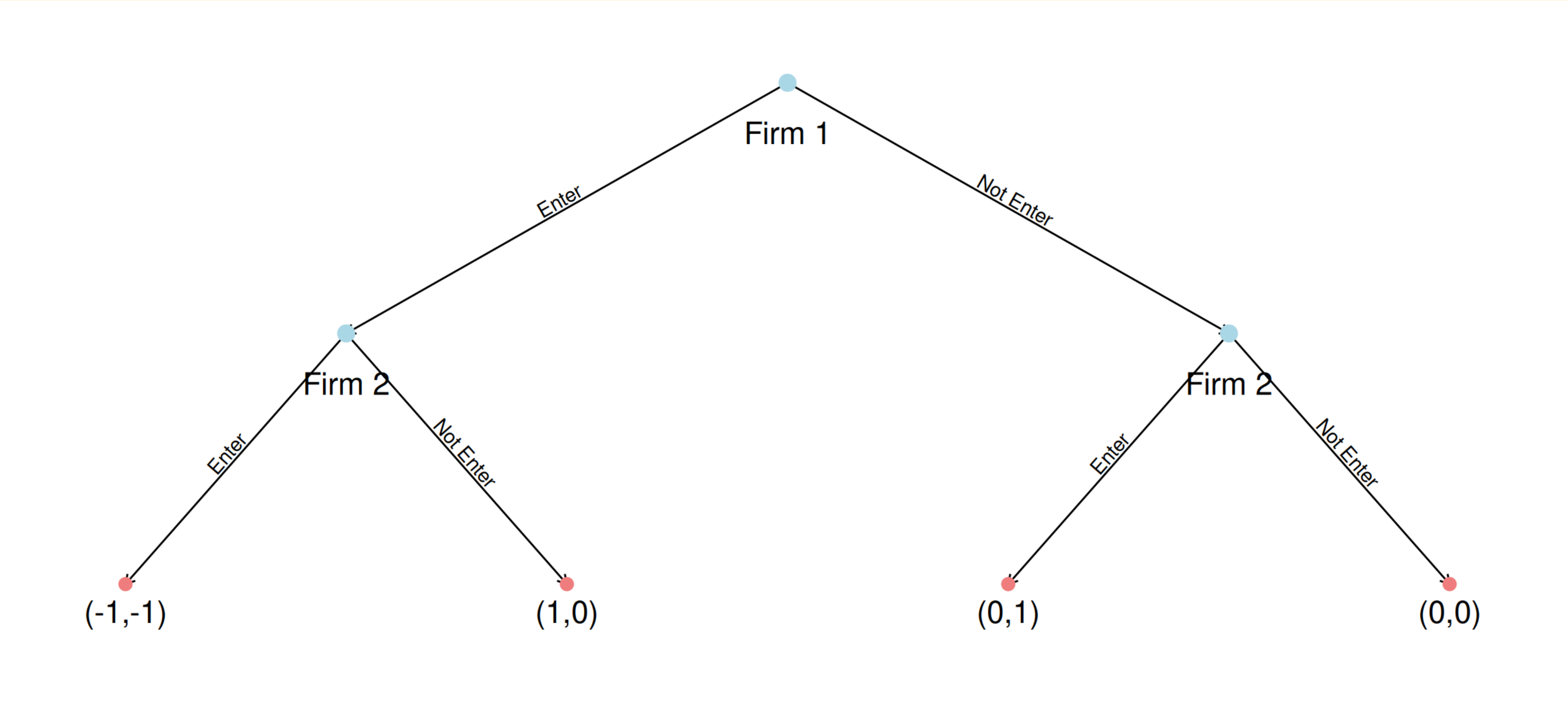


NOW LET'S PLAY THE FIRM ENTRY GAME AS A SEQUENTIAL GAME

We solve sequential games from the end to the start - called backwards induction

What would Firm 2 want to do at each node?

Predicting what Firm 2 is going to do - what should Firm 1 do?



NOW LET'S PLAY THE FIRM ENTRY GAME AS A SEQUENTIAL GAME

In a weird convention, we write our equilibria strategy out for every node from left to right for each layer.

So the equilibrium strategy for this game is:

$$\left(\underbrace{\text{Enter}}_{\text{Firm 1 node}}, \left(\underbrace{\text{Not enter}}_{\text{Firm 2 left node}}, \underbrace{\text{Enter}}_{\text{Firm 2 right node}}\right)\right)$$

We call this the **Subgame perfect equilibrium**

I think it's usually more useful to just think of the equilibrium outcome - ie, Firm 1 enters, Firm 2 does not enter

DRAWING OUT A MORE COMPLEX GAME

Last week I spoke about pharmaceutical companies breaking exclusivity restrictions. Let's map it out as a more complex sequential game.

- First, Firm 1 chooses to enter or not. They should be legally protected by an exclusivity provision.
- Then, Firm 2 chooses to enter or not. If they enter when Firm 1 has already entered, they are breaking the law.
- If both Firm 1 and 2 have entered - Firm 1 can choose to sue Firm 2.
- If Firm 1 acts as a monopoly, it earns \$5. If Firm 2 acts as a monopoly they earn \$3.
- If Firm 1 and 2 compete, they earn \$4 and \$2.
- If Firm 1 sues Firm 2: Firm 1 gets \$0.5 after legal fees, and Firm 2 loses an addition \$1.